ﷺ Online Book Store - SQL Portfolio Project (PostgreSQL)

A complete end-to-end SQL Portfolio Project built using PostgreSQL for Data Analyst Resume. This project analyzes an Online Bookstore's sales, customers, and inventory using 20 real business questions.

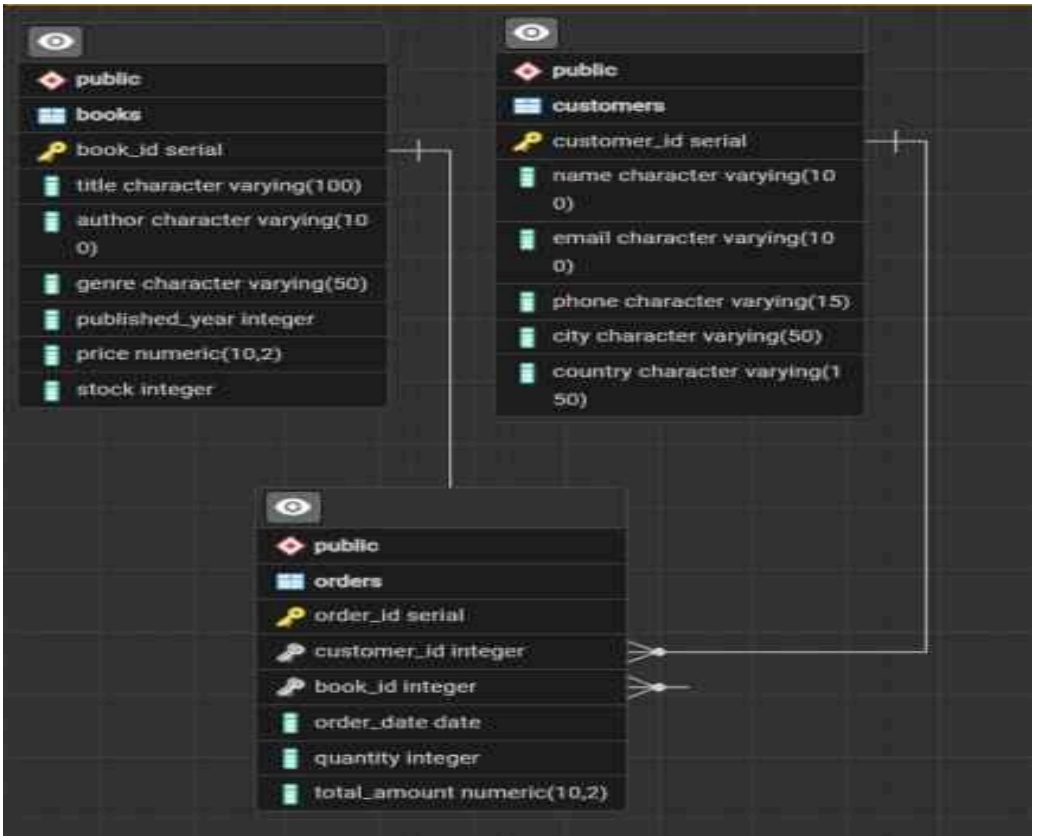
> *Ideal for showcasing SQL skills: JOINS, GROUP BY, Subqueries, Aggregations, and Business Analysis.*

Database & Tables

Database Name: `OnlineBookstore`

Tables Used: `Books.csv` , `Customers.csv` , `Orders.csv`

ER Diagram :



Schema:

```
-- Create Database
CREATE DATABASE OnlineBookstore;

-- Create Tables
DROP TABLE IF EXISTS Books;
CREATE TABLE Books (
        Book_ID SERIAL PRIMARY KEY,
        Title VARCHAR(100),
        Author VARCHAR(100),
        Genre VARCHAR(50),
        Published_Year INT,
        Price NUMERIC(10, 2),
        Stock INT
);

DROP TABLE IF EXISTS customers;
CREATE TABLE Customers (
        Customer_ID SERIAL PRIMARY KEY,
        Name VARCHAR(100),
        Email VARCHAR(100),
        Phone VARCHAR(15),
        City VARCHAR(50),
        Country VARCHAR(150)
);

DROP TABLE IF EXISTS orders;
CREATE TABLE Orders (
        Order_ID SERIAL PRIMARY KEY,
        Customer_ID INT REFERENCES Customers(Customer_ID),
        Book_ID INT REFERENCES Books(Book_ID),
        Order_Date DATE,
        Quantity INT,
        Total_Amount NUMERIC(10, 2)
);
```

?? 20 Business Questions Solved

Basic Filtering

1. Retrieve all books in the "Fiction" genre

2. Find books published after the year 1950

3. List all customers from the Canada

4. List customers who have placed at least 2 orders

5. Find the details of the most expensive book

6. Retrieve all orders where the total amount exceeds $20

7. List all genres available in the Books table

8. Find the book with the lowest stock

9. Show orders placed in November 2023

10. Retrieve the total quantity of books sold by each author

11. Retrieve the total number of books sold for each genre

12. Calculate the total revenue generated from all orders

13. List the cities where customers who spent over $30 are located

14. Find the average price of books in the "Fantasy" genre

15. Find the most frequently ordered book

16. Show all customers who ordered more than 1 quantity of a book

17. Retrieve the total stock of books available

18. Show the top 3 most expensive books of 'Fantasy' Genre

19. Find the top 3 customers who spent the most on orders

20. Calculate the stock remaining after fulfilling all orders

✅ Key Outputs & Screenshots

Q4) Customers with at least 2 orders (139 customers found)

Data Output | Messages | Notifications

	customer_id integer	name character varying (100)	order_count bigint
1	225	Christopher Mccullough	2
2	418	Kiara Blankenship MD	3
3	322	William Cameron	3
4	325	Emily Vargas	4
5	376	Justin Donaldson	2
6	486	Melanie Kelly	2
7	461	Crystal Pierce	3
8	2	Crystal Clements	2
9	149	Jason Robinson	3
10	173	Victoria Dixon	2

Total rows: 139 | Query complete 00:00:00.060

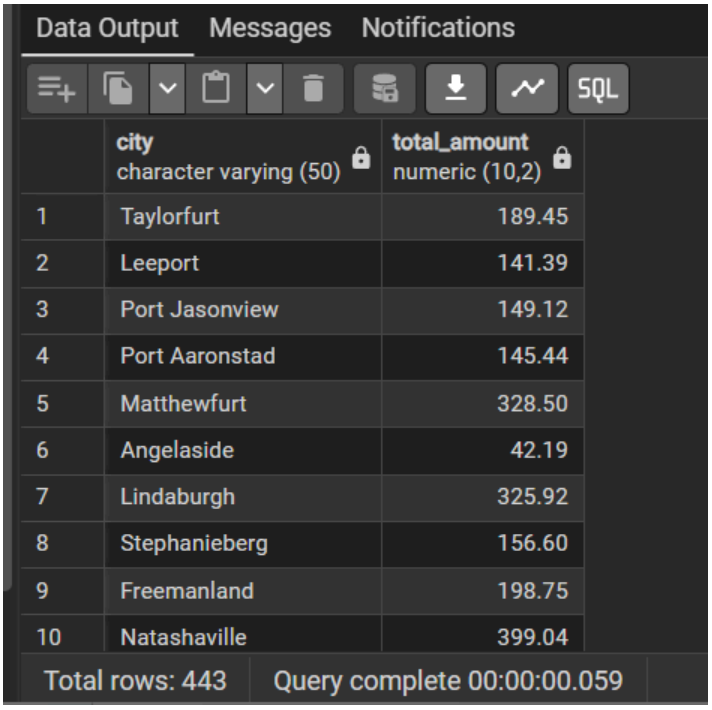
Q7) Total Books Sold by Each Genre

Data Output | Messages | Notifications

	genre character varying (50)	total_books_sold bigint
1	Romance	439
2	Biography	285
3	Mystery	504
4	Fantasy	446
5	Fiction	225
6	Non-Fiction	351
7	Science Fiction	447

Total rows: 7 | Query complete 00:00:00.089

Q12) Cities where customers spent over $30 (443 cities)



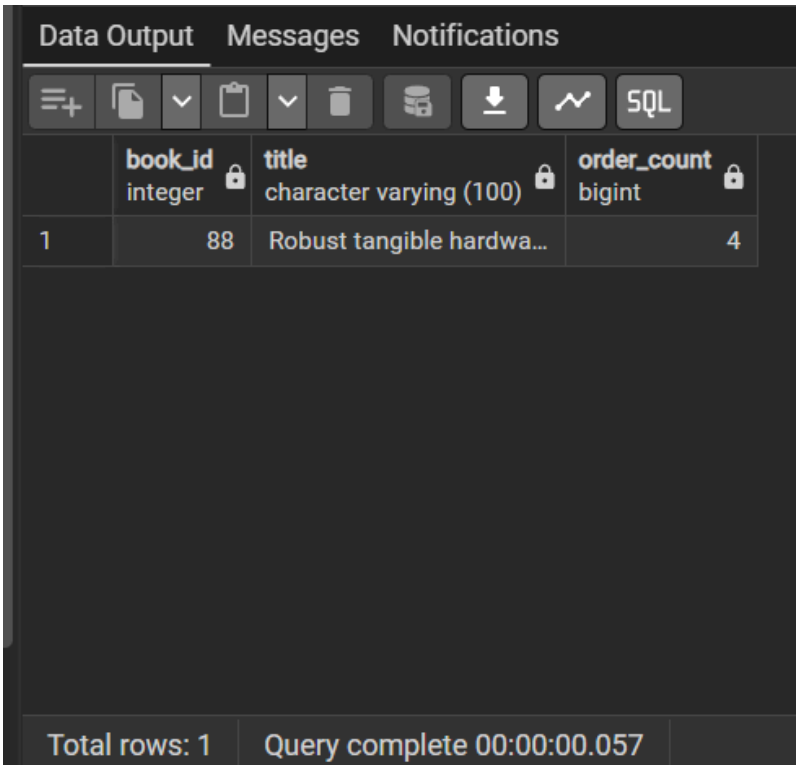
Data Output | Messages | Notifications

SQL

	city character varying (50)	total_amount numeric (10,2)
1	Taylorfurt	189.45
2	Leeport	141.39
3	Port Jasonview	149.12
4	Port Aaronstad	145.44
5	Matthewfurt	328.50
6	Angelaside	42.19
7	Lindaburgh	325.92
8	Stephanieberg	156.60
9	Freemanland	198.75
10	Natashaville	399.04

Total rows: 443 | Query complete 00:00:00.059

Q15) Most Frequently Ordered Book



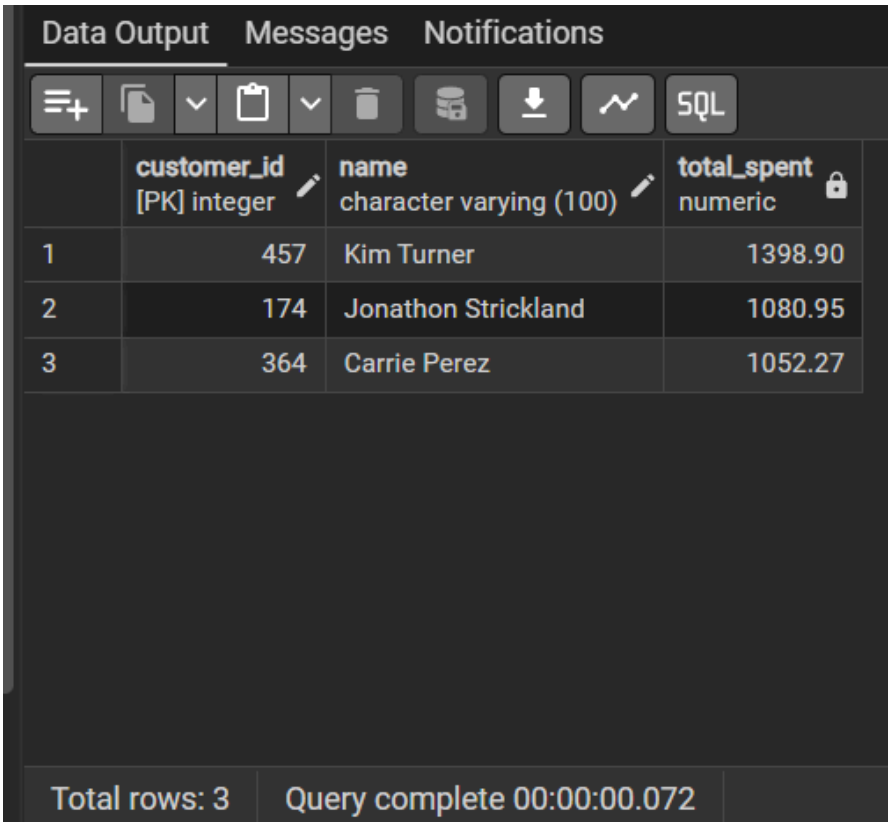
Data Output | Messages | Notifications

SQL

	book_id integer	title character varying (100)	order_count bigint
1	88	Robust tangible hardwa...	4

Total rows: 1 | Query complete 00:00:00.057

Q19) Top 3 Customers Who Spent The Most

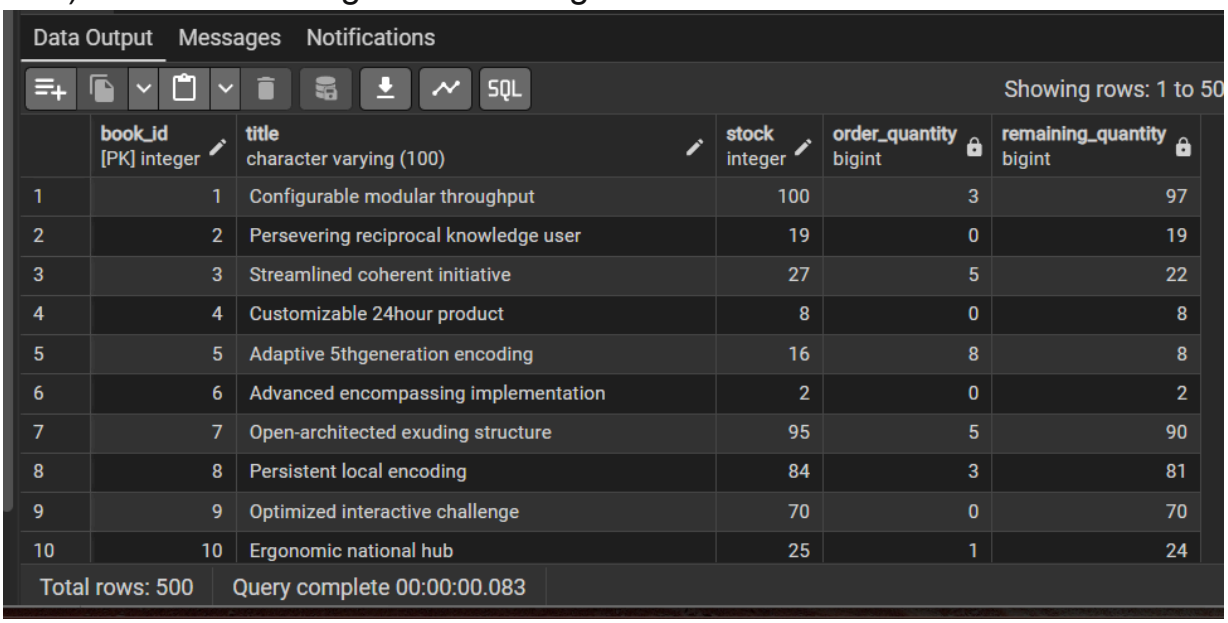


Data Output | Messages | Notifications

	customer_id [PK] integer	name character varying (100)	total_spent numeric
1	457	Kim Turner	1398.90
2	174	Jonathon Strickland	1080.95
3	364	Carrie Perez	1052.27

Total rows: 3 | Query complete 00:00:00.072

Q20) Stock Remaining After Fulfilling All Orders



Data Output | Messages | Notifications

Showing rows: 1 to 50

	book_id [PK] integer	title character varying (100)	stock integer	order_quantity bigint	remaining_quantity bigint
1	1	Configurable modular throughput	100	3	97
2	2	Persevering reciprocal knowledge user	19	0	19
3	3	Streamlined coherent initiative	27	5	22
4	4	Customizable 24hour product	8	0	8
5	5	Adaptive 5thgeneration encoding	16	8	8
6	6	Advanced encompassing implementation	2	0	2
7	7	Open-architected exuding structure	95	5	90
8	8	Persistent local encoding	84	3	81
9	9	Optimized interactive challenge	70	0	70
10	10	Ergonomic national hub	25	1	24

Total rows: 500 | Query complete 00:00:00.083

Skills & Concepts Used :

PostgreSQL & pgAdmin 4

Primary Key, Foreign Key, SERIAL

`WHERE` , `GROUP BY` , `HAVING` , `ORDER BY`

`JOINs` (INNER JOIN), coalesce

Aggregate Functions: `COUNT()` , `SUM()` , `AVG()`

Business Analysis & Data Cleaning

?? How to Run This Project

1. Create Database `OnlineBookstore` in pgAdmin
2. Run the CREATE TABLE queries above
3. Import `Books.csv` , `Customers.csv` , `Orders.csv` using Import/Export Tool (Enable Header)
4. Open `queries.sql` and run each question one by one

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